

13. Derivation of the master equation (2P)

After the Markov approximation, the general form of the master equation for the reduced system density operator is

$$\dot{\rho}_s(t) = -\frac{1}{\hbar^2} \int_{-\infty}^t dt_1 \text{Tr}_B \{ [\tilde{H}_{\text{int}}(t), [\tilde{H}_{\text{int}}(t_1), \tilde{\rho}_s(t) \otimes \rho_B(0)]] \}, \quad (1)$$

where

$$\tilde{H}_{\text{int}}(t) = \hbar \sqrt{\gamma} [c(t) B^\dagger(t) + c^\dagger(t) B(t)]. \quad (2)$$

The double-commutator gives rise to four different contributions and in the lecture we outlined the evaluation of one of them. Repeat the same calculation for the following two contributions

$$\dot{\rho}_s(t) = \frac{1}{\hbar^2} \int_{-\infty}^t dt_1 \left[\text{Tr}_B \{ \tilde{H}_{\text{int}}(t) \tilde{\rho}_s(t) \otimes \rho_B(0) \tilde{H}_{\text{int}}(t_1) \} + \text{Tr}_B \{ \tilde{H}_{\text{int}}(t_1) \tilde{\rho}_s(t) \otimes \rho_B(0) \tilde{H}_{\text{int}}(t) \} \right] \quad (3)$$

and show that this will result in the two 'recycling terms',

$$\dot{\rho}_s = A_- c \rho_s c^\dagger + A_+ c^\dagger \rho_s c, \quad (4)$$

in the final master equation.

$$\dot{\tilde{\rho}}_s(t) = \frac{1}{\hbar^2} \int_{-\infty}^t dt_1 \left[\text{Tr}_B \{ \tilde{H}_{\text{int}}(t) \tilde{\rho}_s(t) \otimes \rho_B(0) \tilde{H}_{\text{int}}(t_1) \} + \text{Tr}_B \{ \tilde{H}_{\text{int}}(t_1) \tilde{\rho}_s(t) \otimes \rho_B(0) \tilde{H}_{\text{int}}(t) \} \right]$$

Coord. transform

$$t_1 = t - \tau \Rightarrow dt_1 = -d\tau$$

$$\text{Tr}_B \{ \tilde{H}_{\text{int}}(t) \tilde{\rho}_s(t) \otimes \rho_B(0) \tilde{H}_{\text{int}}(t-\tau) \} =$$

terms like $B^\dagger(t) S_B(0) B^\dagger(t-\tau)$ and $B(t) S_B(0) B(t-\tau)$ drop (for thermal states) since they do not conserve excitation number

$$S_B = S_B(0)$$

$$= \frac{1}{\hbar^2} \text{Tr}_B \left\{ \left(B^\dagger(t) S_B B(t-\tau) \otimes c(t) S_s(t) c^\dagger(t-\tau) + B(t) S_B B^\dagger(t-\tau) \otimes c^\dagger(t) S_s(t) c(t-\tau) \right) \right\}$$

$$\downarrow \text{all together with } c(t) = e^{-i\omega_s t} c(t-\tau) \\ c^\dagger(t) = e^{i\omega_s t} c^\dagger(t-\tau)$$

$$\dot{\tilde{\rho}}_s(t) = \gamma \int_0^\infty d\tau \left[\langle B(t-\tau) B^\dagger(t) \rangle e^{-i\omega_s \tau} \otimes c S_s(t) c^\dagger + \langle B^\dagger(t-\tau) B(t) \rangle e^{i\omega_s \tau} \otimes c^\dagger S_s(t) c + \langle B(t) B^\dagger(t-\tau) \rangle e^{i\omega_s \tau} \otimes c S_s(t) c^\dagger + \langle B^\dagger(t) B(t-\tau) \rangle e^{-i\omega_s \tau} \otimes c^\dagger S_s(t) c \right]$$

Now use translation invariance in time for the bath

$$\Rightarrow \langle B(t-\tau) B^\dagger(t) \rangle = \langle B(\cdot-\tau) B^\dagger(0) \rangle = \langle B(\tau) B^\dagger(0) \rangle^*$$

$$\text{Held since } = \text{Tr}_B (B(0) B^\dagger(\tau) S_B) = \text{Tr}_B (I S_B B(\tau) B^\dagger(0))^\dagger = (\text{Tr}_B (S_B B(\tau) B^\dagger(0)))^*$$

$$= \gamma \int_0^{\infty} d\tau \left[\langle B(\tau) B^\dagger(0) \rangle^* e^{-i\omega_s \tau} + \langle B(\tau) B^\dagger(0) \rangle e^{i\omega_s \tau} \right] c_{S_5(4)} c^\dagger \\ + \left(\langle B^\dagger(\tau) B(0) \rangle^* e^{i\omega_s \tau} + \langle B^\dagger(\tau) B(0) \rangle e^{-i\omega_s \tau} \right) c^\dagger c_{S_5(4)} c$$

$$= \left(\underbrace{\text{Re}(1_-(\omega_s))}_{=: A_-} \right) c_{S_5(4)} c^\dagger + \left(\underbrace{\text{Re}(1_+(\omega_s))}_{=: A_+} \right) c^\dagger c_{S_5(4)} c$$

with: $l_-(\omega) = 2\gamma \int_0^{\infty} d\tau \langle B(\tau) B^\dagger(0) \rangle e^{i\omega \tau}$

$$l_+(\omega) = 2\gamma \int_0^{\infty} d\tau \langle B^\dagger(\tau) B(0) \rangle e^{-i\omega \tau}$$

14. Stroboscopic master equation (2P)

Consider a cavity QED system as discussed in the lecture in Sec. 3.3, where Rydberg atoms with states $|g\rangle$ and $|e\rangle$ are interacting resonantly with a single microwave mode. During the time of interaction the Hamiltonian is

$$H_{\text{int}} = \frac{\hbar g}{2} (a\sigma_+ + a^\dagger\sigma_-). \quad (5)$$

Assume that at a time t the cavity is an arbitrary pure state $|\psi_c(t)\rangle$ and a Rydberg atom in state $|g\rangle$ is sent toward the cavity. Therefore, right before the interaction the total state of the system is $|\Psi(t)\rangle = |\psi_c(t)\rangle \otimes |g\rangle$.

- (a) Assume that the Rydberg atom is very fast such that the parameter $\varepsilon = \tau g/2 \ll 1$ is small, but finite. Evaluate the state after the interaction,

$$|\Psi(t+\tau)\rangle = e^{-iH_{\text{int}}\tau/\hbar} |\Psi(t)\rangle, \quad (6)$$

up to order ε^2 .

$$\begin{aligned} |\Psi(t+\tau)\rangle &= e^{-iH_{\text{int}}\tau/\hbar} |\Psi(t)\rangle \\ &= \left(1 - i\frac{g\tau}{2}(a\sigma_+ + a^\dagger\sigma_-) - \frac{g^2\tau^2}{8}(a\sigma_+ + a^\dagger\sigma_-)^2 + O(\varepsilon^3)\right) |\psi_c(t)\rangle \otimes |g\rangle \\ &\approx \left(1 - i\varepsilon(a\sigma_+ + a^\dagger\sigma_-) - \frac{\varepsilon^2}{2}(\underbrace{a^2\sigma_+^2}_{\rightarrow 0} + \underbrace{a^{\dagger 2}\sigma_-^2}_{\rightarrow 0} + \underbrace{aa^\dagger\sigma_+\sigma_- + a^\dagger a\sigma_-\sigma_+}_{\rightarrow 0})\right) |\psi_c(t)\rangle \otimes |g\rangle \\ &= |\psi_c(t)\rangle \otimes |g\rangle - i\varepsilon(a|\psi_c(t)\rangle \otimes |e\rangle - \frac{\varepsilon^2}{2}(a^\dagger a|\psi_c(t)\rangle) \otimes |g\rangle \end{aligned}$$

- (b) After the interaction the atom hits a wall and it is lost. Take the trace over the atomic degrees of freedom and calculate the reduced cavity state $\rho_c(t+\tau) = \text{Tr}_{\text{atom}}\{|\Psi(t+\tau)\rangle\langle\Psi(t+\tau)|\}$, again up to order ε^2 .

$$\begin{aligned} \rho_c(t+\tau) &= \text{Tr}_{\text{atom}} \{ |\Psi(t+\tau)\rangle\langle\Psi(t+\tau)| \} \quad \text{For this part } |\psi_c\rangle := |\psi_c(t)\rangle \text{ (only in } \mathbb{C}) \\ &\quad \text{only consider products of terms with same } |g\rangle \text{ or } |e\rangle \text{ state, rest will drop by Tr}_{\text{atom}} \\ &= \text{Tr}_{\text{atom}} \left\{ |\psi_c, g\rangle\langle\psi_c, g| - \varepsilon^2(a|\psi_c\rangle\langle g| \langle\psi_c|a^\dagger\langle g| - \frac{\varepsilon^2}{2}(a^\dagger a|\psi_c\rangle\langle g| \langle\psi_c, g| \right. \\ &\quad \left. - \frac{\varepsilon^2}{2}|\psi_c, g\rangle\langle\psi_c|a^\dagger a\langle g| + O(\varepsilon^3)) \right\} \\ &\approx |\psi_c\rangle\langle\psi_c| - \varepsilon^2 a|\psi_c\rangle\langle\psi_c|a^\dagger - \frac{\varepsilon^2}{2} a^\dagger a|\psi_c\rangle\langle\psi_c| - \frac{\varepsilon^2}{2} |\psi_c\rangle\langle\psi_c|a^\dagger a \end{aligned}$$

(c) This process is repeated and every time interval τ a new atom initialized in state $|g\rangle$ is sent through the cavity. Use the discrete time derivative

$$\dot{\rho}_c(t) = \frac{\rho_c(t+\tau) - \rho_c(t)}{\tau} = \dots \quad (7)$$

to derive an effective master equation for $\rho_c(t)$. Is this master equation of Lindblad form? What is the decay rate?

$$\begin{aligned} \dot{\rho}_c(t) &= \frac{\rho_c(t+\tau) - \rho_c(t)}{\tau} = \frac{\frac{g^2\tau^2}{4} a \rho_c(t) a^\dagger - \frac{g^2\tau^2}{8} [a^\dagger a \rho_c(t) + \rho_c(t) a^\dagger a]}{\tau} \\ &= \frac{g^2\tau}{4} (a \rho_c(t) a^\dagger - \frac{1}{2} \{a^\dagger a, \rho_c(t)\}) \Rightarrow \text{The decay rate is } \frac{g^2\tau}{4} \end{aligned}$$

(d) How would the final master equation change, if the atoms are initialized in state $|e\rangle$ before passing through the cavity?

$$\Rightarrow |\psi_c(t)\rangle = |\psi_c(t)\rangle |e\rangle - i\varepsilon a^\dagger |\psi_c(t)\rangle |g\rangle - \frac{\varepsilon^2}{2} a a^\dagger |\psi_c(t)\rangle |e\rangle$$

(So basically $|g\rangle \leftrightarrow |e\rangle$, $a \leftrightarrow a^\dagger$)

$$\begin{aligned} \Rightarrow \rho_c(t+\tau) &\approx |\psi_c(t)\rangle \langle \psi_c(t)| - \varepsilon^2 a^\dagger |\psi_c(t)\rangle \langle \psi_c(t)| a - \frac{\varepsilon^2}{2} a a^\dagger |\psi_c(t)\rangle \langle \psi_c(t)| \\ &\quad - \frac{\varepsilon}{2} |\psi_c(t)\rangle \langle \psi_c(t)| a a^\dagger \end{aligned}$$

$$\Rightarrow \dot{\rho}_c(t) = \frac{g^2\tau}{4} (a^\dagger \rho_c(t) a - \frac{1}{2} \{a a^\dagger, \rho_c(t)\})$$

Physical interpretation:

If the atoms are in the state $|g\rangle$, the cavity can lose excitations to it
 If ———— $|e\rangle$, the cavity can only receive excitations from it via H_{int}

15. Population inversion in a three-level atom (2P)

Consider a three-level atom with an additional state $|r\rangle$ and a level scheme as illustrated in Fig. 1. The master equation for this system is

$$\dot{\rho} = -i[H, \rho] + \frac{\Gamma}{2}\mathcal{D}[\sigma_{ge}]\rho + \frac{\Gamma_r}{2}\mathcal{D}[\sigma_{er}]\rho, \quad (8)$$

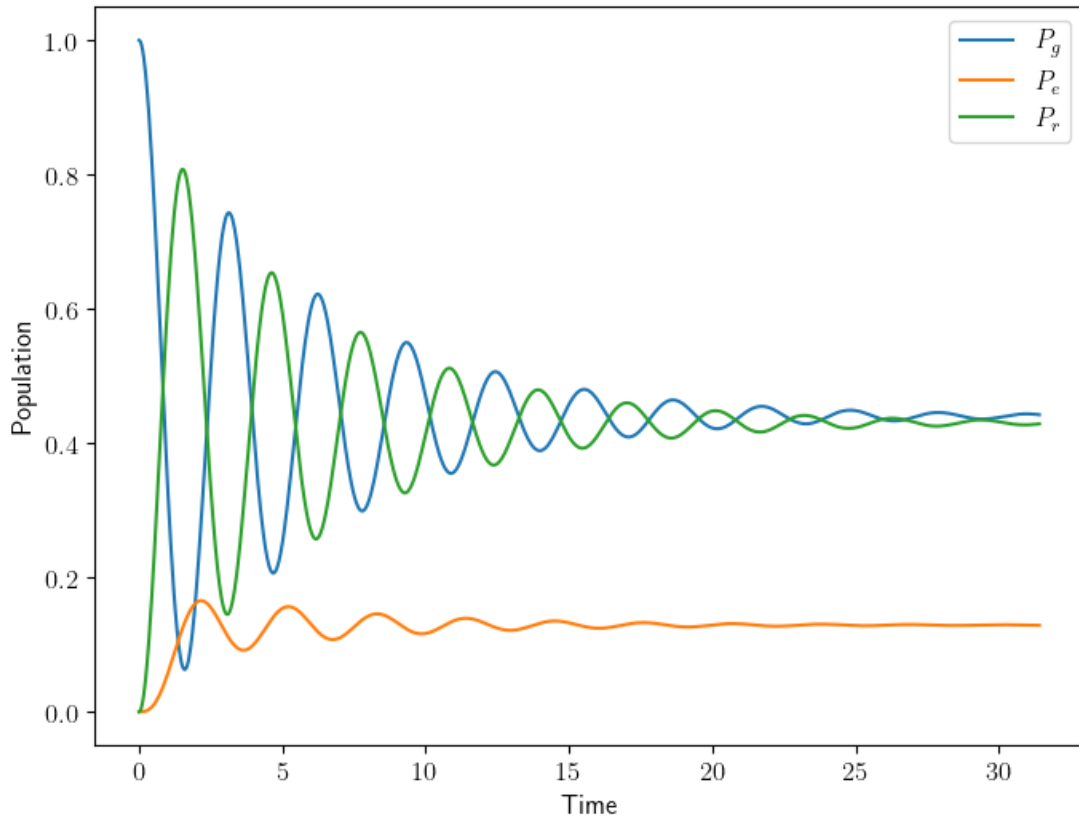
where $\sigma_{ij} := |i\rangle\langle j|$ and

$$H = \frac{\Omega}{2}(\sigma_{gr} + \sigma_{rg}). \quad (9)$$

We have also introduced the short notation

$$\mathcal{D}[c]\rho := 2c\rho c^\dagger - c^\dagger c\rho - \rho c^\dagger c. \quad (10)$$

- (a) Write a Python program to simulate this master equation for a three level system and simulate the dynamics of the populations $p_{\eta=g,e,r}(t) = \langle\sigma_{\eta\eta}\rangle(t)$, starting from the ground state, $\rho(0) = |g\rangle\langle g|$. Plot the results for the specific parameters $\Omega = 2$, $\Gamma = 1$ and $\Gamma_r = 0.3$.



- (b) Evaluate the steady state ρ_{ss} as a function of Ω/Γ and for a ratio between the decay rates $\Gamma_r/\Gamma = 5$. Identify conditions under which a population inversion $p_e(t \rightarrow \infty) > p_g(t \rightarrow \infty)$ can be achieved in the steady state. How can you change the parameters in order to reach an almost ideal inversion, $p_e \rightarrow 1$?

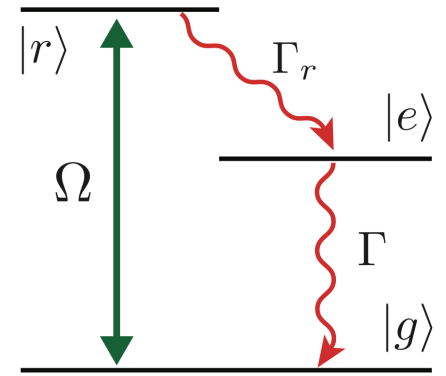
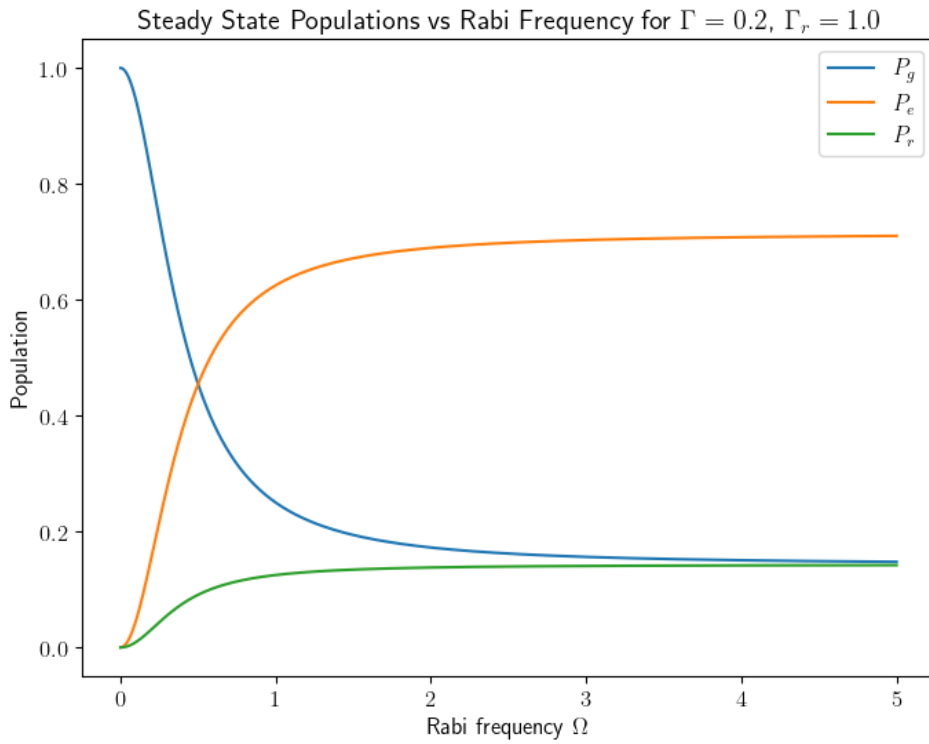


Abbildung 1: Three level atom.

Population inversion happens for $\Gamma < \Gamma_r$ (1) and also $\Gamma < \Omega$ (2)

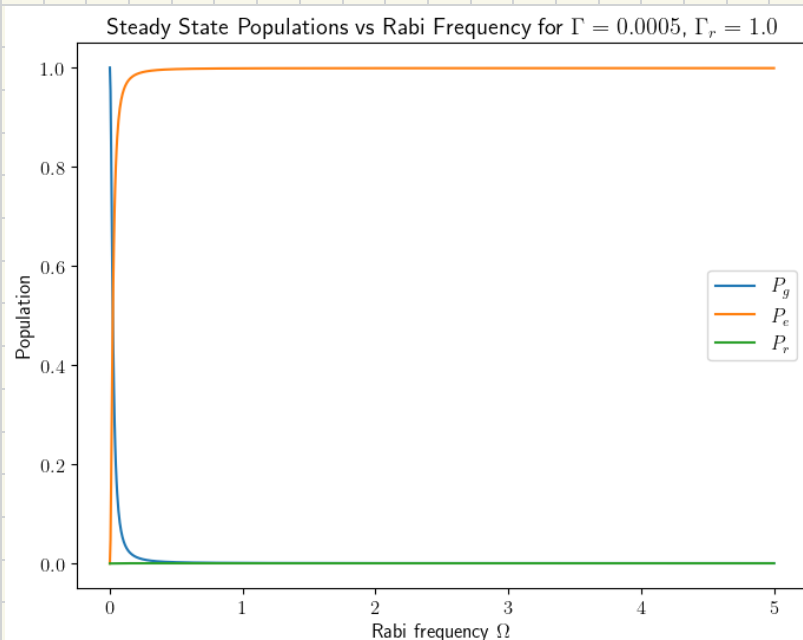
(1) Will lead to the fact that most of the excited population (in $|r\rangle, |e\rangle$) will be in $|e\rangle$

(2) $\Rightarrow$ Excited population will accumulate faster, than it will decay to ground $\Rightarrow$ in the steady state more population will be excited

Rate Eqs.: $\dot{p}_e \stackrel{!}{=} 0 = -\Gamma p_e + \Gamma_r p_r$
 $\dot{p}_g \stackrel{!}{=} 0 = -\alpha \Omega (p_g - p_r) + \Gamma p_e \stackrel{!}{=} 0$
maybe $\alpha = 1$ not sure
 $\dot{p}_r \stackrel{!}{=} 0 = \alpha \Omega (p_g - p_r) - \Gamma_r p_r$

$\Rightarrow \Gamma p_e - \Gamma_r p_r = 0 \Rightarrow p_e = \frac{\Gamma_r}{\Gamma} p_r$

but also $p_g = \frac{\alpha \Omega + \Gamma_r}{\alpha \Omega} p_r \Rightarrow \frac{p_e}{p_g} = \frac{\Gamma_r}{\Gamma} \frac{\alpha \Omega}{\alpha \Omega + \Gamma_r} \quad (\stackrel{!}{>} 1 \text{ for pop. inversion})$
(1) (2)



For very small Γ_r

$\Rightarrow p_{e,ss} = 0.999$